

Infer Low-Access Share

- 45.2% of residents in low-income area live in food desert
- Identify key predictors of the proportion of low-access residents
- Train/test sets split 80/20
- Forward stepwise linear regression, 10FCV PCR

Linear Regression Results

Predictor	Estimate	SE	t
Urban flag	.5315	.0055	95.866
Housing Units w/ Vehicles	-.0005	9.3×10^{-6}	49.029
Hispanic population	-5.3×10^{-5}	2.6×10^{-6}	20.090
Asian population	-9.7×10^{-5}	7.9×10^{-6}	12.228
Senior population	9.4×10^{-5}	9.2×10^{-6}	10.247
Kids population	6.0×10^{-5}	8.5×10^{-6}	7.020
Median family income	-9.9×10^{-7}	1.6×10^{-7}	6.205
Black population	3.0×10^{-5}	5.1×10^{-6}	5.901
White population	2.2×10^{-5}	4.9×10^{-6}	4.530
Poverty rate	.0010	.0002	4.472
Population	-8.7×10^{-6}	5.6×10^{-6}	1.549

References

- Ek, C., Hébert, J. R., Friedman, D. B., & Porter, D. E. (2025). Climate change, racism, and food insecurity: Cyclical impacts of stressors exacerbate health disparities. *Journal of racial and ethnic health disparities*, 12(6), 4082-4093.
- Ver Ploeg, M. et al. (2012). *Access to Affordable and Nutritious Food: Measuring and Understanding Food Deserts and Their Consequences*. USDA Economic Research Service Report to Congress.

Inferential modeling of food desert risk across U.S. census tracts

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Summary

Among low-income census tracts, 45.2% of residents lack access to a supermarket within either half a mile (for urban areas) or ten miles (for rural areas). Whether a tract is urban or not, the number of housing units without a vehicle within the tract, and the Hispanic/Latino population count for the tract best explained variation in this continuous access shortfall.

We estimate that census tracts can reliably and sensitively be classified as a USDA-defined food desert using poverty rate, vehicle share, and population density.

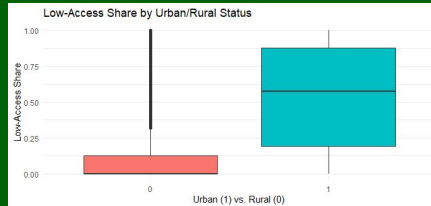
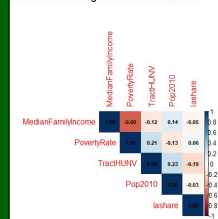
Regression Model Comparison

Model	R ²	RMSE	p
Forward Stepwise	.3368	.3073	11
PCR	.3366	.3073	11

Classification Model Comparison

Model	Threshold	Sens	Bal. Acc.	p
Lasso (1 SE)	.14	.9552	.7199	3
Logit	.1	.9527	.6880	3

Correlation Among Candidate Predictors



Data Overview

- 72,531 tracts
- 147 variables
- Data sourced from USDA Food Access Research Atlas

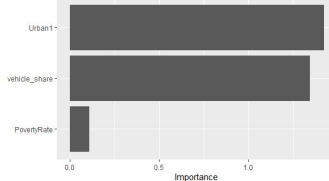
Infer Food Desert

- Food desert classified as low-income, low-access tracts at half-mile (urban) or 10-mile (rural) distance
- Identify key predictors of the food deserts
- Train/test sets split 80/20
- 10FCV LASSO, logistic regression

LASSO (1 SE) Coefficients

Predictor	Estimate
Urban flag	1.424
Housing units w/ vehicles (%)	1.345
Poverty rate	0.108

LASSO Variable Importance



Logit ROC Curve

